

Pain Management

Pain management requires a combination of conservative measures, medications, and appropriate wound care. Muscle relaxants and physical and occupational therapy may be helpful to decrease muscle spasm in the area of ulceration. Transcutaneous electrical nerve stimulation may help relieve acute and chronic pain. The use of topical anesthetics such as lidocaine-prilocaine cream on wounds 30 minutes before débridement significantly reduces pain associated with wound care procedures.

Nonopioid analgesics (aspirin, other nonsteroidal anti-inflammatory drugs) are first-line systemic therapy, followed by stronger medications such as opioids. Adjuvant medications such as tricyclic antidepressants can be used to enhance analgesia. They can also improve depression in some chronic pain states and have sedative, sleep-enhancing qualities.

Nutritional Management

Screening for nutritional deficiencies is an important part of assessment and management of pressure ulcers. Adequate patient dietary intake of protein is important in healing pressure ulcers. A reasonable protein amount is between 1.2 and $1.5\,\mathrm{g/kg/day}$, and caloric intake should be between 30 and $35\,\mathrm{kcal/kg/day}$. Vitamin C or zinc deficiency can have negative effects on wound healing, but supplementation is useful only when deficiency is confirmed or suspected. If dietary intake continues to be inadequate, oral or parenteral nutritional supplementation may be required. Markers of nutritional deficiency include a serum albumin level of less than $3.5\,\mathrm{mg/dL}$ and unintentional weight loss of more than 15 percent of baseline. Patients should be kept adequately hydrated.

Surgery

The long-term efficacy of surgical repair of pressure ulcers, especially in older patients, has been questioned because of high recurrence rates and mortality. The decision to undertake surgical repair should be made with full knowledge of the risks and benefits, surgical and nonsurgical alternatives, and complications associated with each procedure. Surgical techniques include skin grafting; creation of cutaneous, muscle, and fasciocutaneous flaps or pedicled flaps; microvascular free tissue transfer reconstruction; and simple direct suture, depending on the site, number, and grade of lesions, and the patient's clinical condition. In general, surgery is often required for stage IV ulcers, whereas more superficial ulcers may heal with conservative therapy.

Adjunctive Therapies

Adjunctive therapies have been used in case studies, but more data are needed to determine the effectiveness of these therapies. These therapies are outlined in the electronic version of this book.

Whenever possible, other factors that impair healing—such as smoking, hypovolemia, cold exposure, chronic diseases (including diabetes mellitus, systemic arterial hypertension, anemia, and vascular alterations), and the use of medications such as corticosteroids—should be limited or controlled.

Prevention

Preventive actions can decrease the patient's risk of developing pressure ulcers. These steps can also keep pressure ulcers from getting worse. Preventive measures are listed in Table 99-6.

TABLE 99-6 Preventive Measures

- Maintain personal hygiene.
- Assure adequate nutrition and hydration.
- Evaluate and manage urinary and fecal incontinence.
- Position to alleviate pressure over bony prominences and shearing forces over heels, elbows, base of head, and ears.
- Reposition every 2 hours when in bed and every hour when in a chair; if alert and capable, the patient should be taught to shift weight every 15 minutes while in a chair.
- Use appropriate positioning devices and foam padding; do not use donut-shaped devices.
- Avoid placing the patient on the trochanters or directly on the wound.
- Maintain the lowest head elevation possible.
- Use lifting devices such as draw sheets or a trapeze.
- Prevent contractures.
- Do not massage reddened areas over bony prominences.